System Validation Report

Project: Online Bookstore — Team 12

Lead QA / Tester: D Sai Karthik (PES2UG24CS141)

Testing Cycle: Sprint 2 Quality Assurance & Final Acceptance

Date: 06 November 2026

Document Reference: `Online_Bookstore_SRS_Team12_FINAL_FIXED.md` (Section 4)

Status: 100% VALIDATED & ACCEPTED

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1. Executive Summary

System Validation ensures that the built application satisfies all specified user needs, business requirements, and operational workflows as outlined in the Software Requirements Specification (SRS).

This report provides complete functional acceptance verification for all **23 Functional Requirements (BOOK-F-001 through BOOK-F-023)** across all three system roles: **Buyer**, **Seller**, and **Administrator**.

High-Level Validation Results

- **Total Functional Requirements (FRs) Evaluated: 23 / 23**
- **Total Requirements Fully Validated: 23 (100%)**
- **Failed / Incomplete Requirements: 0**
- **System Acceptance Decision: ACCEPTED FOR RELEASE**

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2. Functional Requirements Traceability & Validation Matrix

Req ID	Requirement Description	User Persona	Validation Method	Acceptance Criteria Result	Final Status
BOOK-F-001	Visitor registration with unique email and password complexity	Buyer / Seller	Jest + Vitest + Browser	Rejects duplicate emails; validates password complexity; stores salted hash	**VALIDATED**
BOOK-F-002	User authentication with email/password and JWT session	All Users	Jest + Vitest + Browser	Correct credentials issue token; wrong credentials denied; 5 failures lock account	**VALIDATED**
BOOK-F-003	Password reset via secure time-limited token	All Users	Jest + Vitest	Cryptographically random single-use token expires after 30 minutes	**VALIDATED**
BOOK-F-004	Profile editing and multiple shipping address management	Buyer / Seller	Jest + Vitest + Browser	User can view/edit profile and manage multiple saved shipping addresses	**VALIDATED**

Req ID	Requirement Description	User Persona	Validation Method	Acceptance Criteria Result	Final Status
BOOK-F-005	Guest catalogue browsing by category/genre	Guest / Buyer	Vitest + Browser	Category filter lists active books without requiring authentication	**VALIDATED**
BOOK-F-006	Keyword search across title, author, and ISBN with pagination	Guest / Buyer	Jest + Vitest + Browser	Search bar returns matching active listings with limit and page controls	**VALIDATED**
BOOK-F-007	Book detail page displaying price, condition, seller, stock	Guest / Buyer	Vitest + Browser	Detailed page renders all attributes, stock indicators, and seller details	**VALIDATED**
BOOK-F-008	Search and browse filtering and sorting (price, rating, availability)	Guest / Buyer	Jest + Vitest + Browser	Sorts ascending/descending and filters out-of-stock titles correctly	**VALIDATED**
BOOK-F-009	Verified seller listing creation with ISBN format validation	Seller	Jest + Vitest + Browser	Seller creates listing with ISBN-10/13 validation; listing appears immediately	**VALIDATED**
BOOK-F-010	Seller editing of existing listing details (price, stock, condition)	Seller	Jest + Vitest + Browser	Seller can modify own listing; unauthorized users rejected (HTTP 403)	**VALIDATED**
BOOK-F-011	Seller deactivation of listing & duplicate ISBN flagging	Seller / Admin	Jest + Vitest	Seller soft-deactivates listing; duplicate ISBN auto-flagged for moderation	**VALIDATED**
BOOK-F-012	Seller dashboard displaying active listings, orders, earnings	Seller	Vitest + Browser	Dashboard displays accurate aggregates reconciled with database	**VALIDATED**
BOOK-F-013	Persistent shopping cart surviving user sessions	Buyer	Jest + Vitest + Browser	Cart items and quantities persist across browser refreshes and login/logout	**VALIDATED**
BOOK-F-014	Coupon discount code validation and application at checkout	Buyer	Jest + Vitest + Browser	Valid codes (`SAVE10`) apply discounts; expired/below-minimum codes rejected	**VALIDATED**
BOOK-F-015	Shipping address selection or creation during checkout	Buyer	Vitest + Browser	User selects default address or enters new address dynamically during checkout	**VALIDATED**

Req ID	Requirement Description	User Persona	Validation Method	Acceptance Criteria Result	Final Status
BOOK-F-016	Integrated payment gateway checkout (Razorpay mock)	Buyer	Jest + Vitest + Browser	Valid payment confirms order, decrements stock, and clears persistent cart	**VALIDATED**
BOOK-F-017	Order confirmation and downloadable PDF invoice generation	Buyer	Jest + Supertest + Browser	Generated PDF invoice matches line items, discount, tax, and order totals	**VALIDATED**
BOOK-F-018	Order delivery lifecycle tracking with status progression	Buyer / Admin	Jest + Vitest + Webhook	Displays stages (`Placed` -> `Shipped` -> `Out for Delivery` -> `Delivered`)	**VALIDATED**
BOOK-F-019	Order cancellation and return requests within 7-day policy	Buyer	Jest + Vitest + Browser	Cancellation allowed when Placed; return accepted within 7-day window	**VALIDATED**
BOOK-F-020	Verified-purchaser rating and review submission	Buyer	Jest + Vitest + Browser	Only users with delivered order can review; non-purchasers rejected (403)	**VALIDATED**
BOOK-F-021	Automatic aggregate rating recalculation	All Users	Jest + Vitest	Weighted average rating recalculates immediately upon new review submission	**VALIDATED**
BOOK-F-022	Administrator listing and report moderation queue	Admin	Jest + Vitest + Browser	Admin approves/rejects flagged listings; records audit trail in `AuditLog`	**VALIDATED**
BOOK-F-023	Administrator account suspension and sales analytics	Admin	Jest + Vitest + Browser	Admin suspends malicious accounts; sales revenue analytics reconcile with orders	**VALIDATED**

3. End-to-End User Journey Validation

System acceptance was confirmed through manual browser walkthroughs covering complete lifecycle workflows:

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[Guest Browse & Search] ➔ [Registration & Login] ➔ [Cart & Coupon] ➔ [Payment Checkout] ➔ [Order Tracking & PDF Invoice]

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3.1 Buyer Experience

1. **Catalog Exploration:** Browsed catalogue at `http://localhost:5173/`, selected category "Computer Science", and searched for "Design Patterns".
2. **Account Onboarding:** Successfully registered a buyer profile; received automatic session login.
3. **Cart Management:** Added 2 copies of a selected book; entered coupon `SAVE10` and verified a 10% reduction in cart total.
4. **Checkout:** Selected saved address, initiated simulated payment, confirmed order, and verified that cart was emptied.
5. **Post-Purchase:** Navigated to Orders page, verified "Placed" status, downloaded generated PDF invoice, and confirmed order line items matched checkout totals.

3.2 Seller Experience

1. **Listing Management:** Authenticated seller navigated to `/seller/create`, filled in title, author, price (■499), stock (15), and valid ISBN-13.
2. **Dashboard Review:** Verified that the newly created listing appeared on the Seller Dashboard (`/seller/dashboard`) and in public catalogue search.
3. **Inventory Update:** Updated stock count from 15 to 20; verified immediate reflection on book detail view.

3.3 Administrator Experience

1. **Moderation Queue:** Logged in with admin account, navigated to `/admin`, and inspected flagged duplicate listing report.
2. **Action Execution:** Approved resolution action, confirming that the duplicate listing was removed from public browse and recorded in `AuditLog`.
3. **Analytics Dashboard:** Verified aggregate metrics for total platform revenue, active seller count, and daily order volume.

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4. System Validation Sign-Off

The Online Bookstore application (Team 12) complies with all requirements stipulated in the Software Requirements Specification (SRS) and passes all functional acceptance criteria.

Signed Off By:

D Sai Karthik (PES2UG24CS141) — Lead QA Tester

Gautam — Development Lead

Team 12, Online Bookstore Project